The Kernel as an Open System:

Maxwell's Demon, Landauer's Bound, and the Lindblad Structure of a Running Viable System

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Skeleton — July 2, 2026

Abstract

[TO WRITE — one paragraph per lens: (i) a working cybernetic kernel treated as a thermodynamically open system; (ii) an empirical Maxwell demon on real dynamics (ensemble cost 0.31 bits vs. 500 interventions, 1100× discrepancy); (iii) the Landauer accounting that restores the Second Law; (iv) the Lindblad master-equation reading: the $\varphi^{\uparrow}/\varphi^{\downarrow}$ channel pair as measurement-and-drive apparatus, the attractor as engineered steady-state. Audience: complex systems / open quantum systems community (Fierro UV, SFI).]

1 Introduction: Thermodynamic Information as a Common Language

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2 The Shannon–Boltzmann Bridge

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3 The VSF Kernel as an Open System

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4 Experiment: Maxwell's Demon on Real Dynamics

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5 Landauer's Bound and the Restoration of the Second Law

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6 The Lindblad Connection

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7 The Quantum Computing Limit: Decoherence as the Unavoidable Demon

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8 Discussion

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9 Open Questions and Future Experiments

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